Capstone 6 – Before vs After Study Analysis

1. What is being compared

	Before	After
Study method	Read the raw code and the textbook explanation only	Read AI explanation + run the code in Colab + use mini-guide and `check_answer()`
Material	The same 10 programs	The same 10 programs
Test	Pre-test: 10 questions (document 5), no help	Post-test: the same 10 questions in a different order, a week later

2. Procedure

1. Pick 3–5 classmates (or yourself on two separate days) who are Python beginners.
2. **Before:** each person studies the programs without AI for 20 minutes, then takes the quiz with `grade_quiz()`. Record score and time.
3. **After:** same people study with the AI explanations, the mini-guide and the Colab checker for 20 minutes, then retake the quiz.
4. Record how many AI claims each person caught as wrong.

3. Results table (fill with your real data)

Participant	Before score /10	After score /10	Change	Wrong AI claims they caught	Confidence before (1–5)	Confidence after (1–5)
1						
2						
3						
Average						

Average change = (average after) − (average before). Add a bar chart in Google Sheets or Gamma.

4. Questions most likely to improve

The trap questions: Q1 (range end), Q4 (insert), Q9 (shared default). These are exactly where wrong AI claims clustered in the error log.

5. Analysis template

- **Result:** Average score moved from __ to __ (change of __ points).
- **Where it helped:** __ (for example trace tables from prompt PR2).
- **Where it did not help:** __ (for example learners who trusted a wrong AI claim).
- **Effect of verification:** compare people who ran the code with people who only read the AI text.
- **Caution:** with 3–5 people this is an observation, not proof. Learning between tests and repeating the same questions can raise scores on their own.

6. Fairness checks

- Use different question order in the post-test.
- Record whether anyone had prior Python experience.
- Report any participant whose score dropped; do not hide it.

7. Conclusion template

"Using AI explanations together with running the code and an automatic checker changed average quiz scores from __ to __. The biggest gains were on __ questions. AI explanations alone were not enough because __ of __ claims were wrong."